REDUCTION MODULO p AND A BOUND ON THE TORSION

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The aim is a `Magma` script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 1 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, section 2 turns this into two bounds, and section 3 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant Δ.
- The torsion subgroup $T \coloneqq E(\mathbb{Q})_{\mathrm{tors}}$.
- A prime p is good if $p \nmid \Delta$. Then reducing the equation modulo p gives an elliptic curve E_p over $\mathbb{F}_p$.
- For an abelian group G, $G[m]$ is the subgroup killed by m.
- A finite abelian group G is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leqslant n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

How to use this. Allow 20 minutes for §1 and 30 for §2. Section 3 is for home. Every `Magma` block runs in seconds in the free online calculator. Paste them in order, since later blocks use earlier functions.

1. Reduction is injective on torsion

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 1.6 is from Application VII.3.5 there.

In the following definition and results, p is a good prime for E.

Definition 1.1 (Reduction modulo p)**.** We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

Proposition 1.2 (Minimality [Sil09, Remark VII.1.1])**.** *The equation is minimal at p, because $v_p(\Delta) = 0 < 12$. So E_p is the reduction of E modulo p.*

Proposition 1.3 (Reduction is a homomorphism [Sil09, Proposition VII.2.1 and the discussion after it])**.** *The reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of O and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$.*

Proposition 1.4 (Torsion prime to p [Sil09, Proposition VII.3.1])**.** *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$.*

Theorem 1.5 (Torsion of p-power order [Sil09, Theorem VII.3.4(b)])**.** *Let $P \in E(\mathbb{Q}_p)$ have exact order p^n with $n \geqslant 1$, and let $r \coloneqq \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r}x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$.*

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Problem 1.6 (Injectivity of reduction on torsion)**.**
(1) Let $p \geqslant 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$$

is injective on T. In particular T is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.[a]
(2) The prime 2 is different: even if $p = 2$ is good, $T \to E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E\colon y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);          // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));    // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8];  2*P eq E!0;              // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 1.4 and Theorem 1.5 fails, and what does Proposition 1.4 still give at $p = 2$?

[a] *Hint:* a nonzero torsion point in $E_1(\mathbb{Q}_p)$ has a multiple of prime order in $E_1(\mathbb{Q}_p)$.

2. Two bounds

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

Problem 2.1 (The order bound)**.**
The first piece of the script computes the greatest common divisor b of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leqslant N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);          // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
E2 := EllipticCurve([0, 3]);                    // y^2 = x^3 + 3
Discriminant(E2);                               // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]]; // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                             // 1
```

(1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.
(2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);      // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);             // -611 -324710451
IsIsomorphic(E1, E4);                           // true
```

```
OrderBound(E1, 7), OrderBound(E4, 7);            // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

Corollary 2.2 (The m-torsion [Sil09, Corollary III.6.4]). *Let E be an elliptic curve over a field K, and $m \geqslant 1$. If $\operatorname{char} K \nmid m$, then $E[m](\overline{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$, then $E[p^e](\overline{K})$ is cyclic for every $e \geqslant 1$.*

Problem 2.3 (The structure bound).
The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns $[\,]$ when G is trivial:

```
E3 := EllipticCurve([1, 0]);                     // y^2 = x^3 + x
Discriminant(E3);                                // -64
OrderBound(E3, 30);                              // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [ [ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

(1) Let G be a finite subgroup of $E(\overline{K})$, for an elliptic curve E over a field K. Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of G. Show that $\#E[n_1(G)](\overline{K}) = n_1(G)^2$, and that an integer $m \geqslant 1$ with $\#E[m](\overline{K}) = m^2$ satisfies $E[m](\overline{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.[a]
(2) Let b_1 and b_2 be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leqslant N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0;  b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30);                          // [ 1, 2 ]
P := E3![0, 0];  2*P eq E3!0;                    // true
```

Conclude that $T = \{O, (0,0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does b stay at 4 however large N is?
(3) Show that a rational point P is a torsion point if and only if $b_2 P = O$, and that rational torsion points generating a subgroup of order $b_1 b_2$ generate T.

[a] *Hint:* for a prime ℓ, the group $G[\ell]$ lies in $E[\ell](\overline{K})$. Use Corollary 2.2.

3. How good are the bounds? The LMFDB

The L-functions and modular forms database [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

Proposition 3.1 (Reduction of isogenies [Mil20, proof of Theorem V.4.1])**.** *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime p of both equations.*

Proposition 3.2 (Isogenous curves over $\mathbb{F}_p$ [Sil09, Exercise 5.4(a)])**.** *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points.*

Problem 3.3 (Testing the bounds)**.**
(1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, a-invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7o LMFDB %-9o order bound %-3o structure bound %o\n",
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1   LMFDB [ 1 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a2   LMFDB [ 2 ]     order bound 6   structure bound [ 1, 6 ]
// 19.a2   LMFDB [ 3 ]     order bound 3   structure bound [ 1, 3 ]
// 15.a4   LMFDB [ 4 ]     order bound 8   structure bound [ 1, 4 ]
// 11.a2   LMFDB [ 5 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a6   LMFDB [ 6 ]     order bound 6   structure bound [ 1, 6 ]
// 26.b2   LMFDB [ 7 ]     order bound 7   structure bound [ 1, 7 ]
// 15.a8   LMFDB [ 8 ]     order bound 8   structure bound [ 1, 8 ]
// 54.b2   LMFDB [ 9 ]     order bound 9   structure bound [ 1, 9 ]
// 66.c3   LMFDB [ 10 ]    order bound 10  structure bound [ 1, 10 ]
// 90.c7   LMFDB [ 12 ]    order bound 12  structure bound [ 1, 12 ]
// 15.a2   LMFDB [ 2, 2 ]  order bound 8   structure bound [ 2, 4 ]
// 15.a5   LMFDB [ 2, 4 ]  order bound 8   structure bound [ 2, 4 ]
// 30.a6   LMFDB [ 2, 6 ]  order bound 12  structure bound [ 2, 6 ]
// 210.e6  LMFDB [ 2, 8 ]  order bound 16  structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?
(2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]);        // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]);          // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
```

```
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p)))),
     Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 3.1 and 3.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.
(3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]);  P := E![5, -2];        // 15.a4
[k*P eq E!0 : k in [1..4]];                        // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?
(4) Find these curves yourself: on https://www.lmfdb.org/EllipticCurve/Q/, search by "torsion structure", such as `[2,4]`, and sort by conductor. On a curve's page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

Remark 3.4 (Beyond the local theory)**.** Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2 P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\text{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

References

[LMF25] The LMFDB Collaboration, *The L-functions and modular forms database*, 2025, https://www.lmfdb.org. [Online; accessed 21 July 2025]. ↑

[Mil20] James S. Milne, *Elliptic curves*, Second, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2020. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑